Evaluation Report on TransE and TransR Models

Problem Statement

1-hop Question-Answering and Similar Fact Retrieval in Knowledge Graph:
The objective of this assignment is to develop a custom implementation of two knowledge graph embedding techniques—TransE and TransR— for the task of 1-hop question-answering and design a model for similar fact retrieval to identify and rank the top-5 most similar facts to a given fact.

Dataset Description

The Nations dataset contains 14 entities and 55 relations with 1992 triples describing country relationships. The Kinships dataset describes relationships among members of an Australian tribe
with 10,686 triples, including 104 entities and 26 relationship types.

Implementation Details

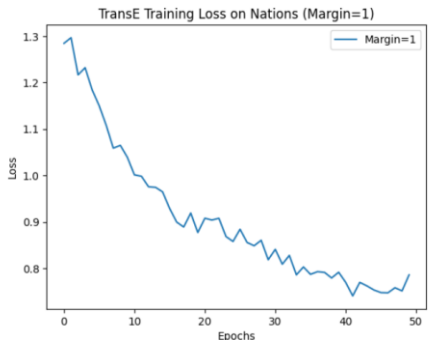
Embedding Dimension: 100,
Margin: Varying 1.0 to 5.0 for Nations, 1.0 for Kinships.
Optimizer: Adam, learning rate 0.001, 50 epochs. Margin-based ranking loss, Bernoulli negative sampling.

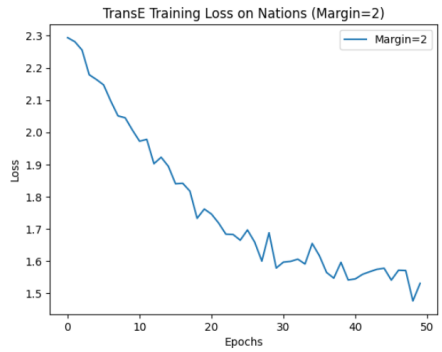
Evaluation Results for Nations Dataset

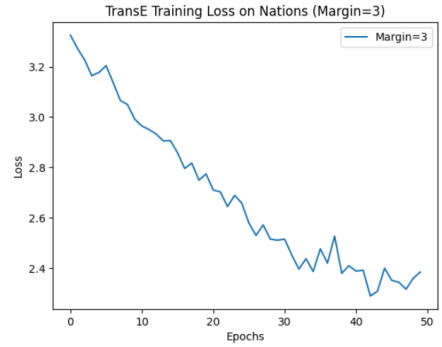
TransE Performance on Nations Dataset

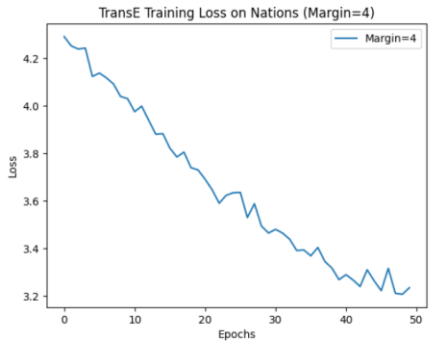
Margin	Mean Rank (Head)	Hits@10 (Head)	Mean Rank (Tail)	Hits@10 (Tail)
1	7.90	0.72	6.96	0.78
2	8.00	0.69	6.85	0.79
3	**7.82**	**0.70**	**6.97**	**0.81**
4	7.76	0.68	6.97	0.79
5	7.84	0.70	6.99	0.80

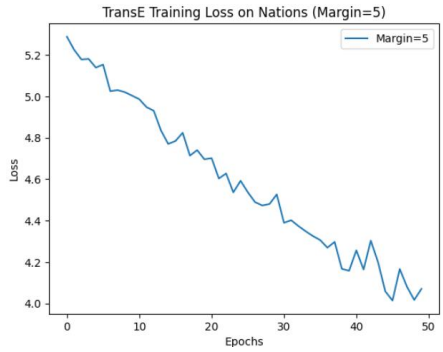
Training Loss on Nations dataset for margin 1 to 5 for TransE







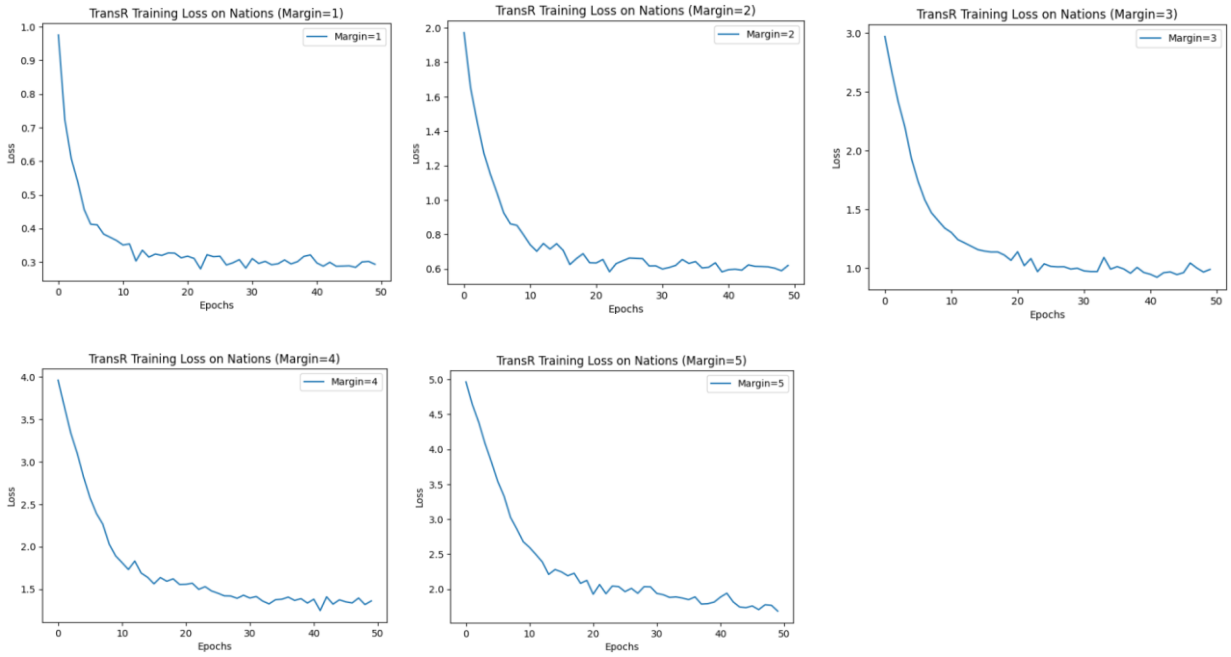




TransR Performance on Nations Dataset

Margin	Mean Rank (Head)	Hits@10 (Head)	Mean Rank (Tail)	Hits@10 (Tail)
1	8.20	0.70	7.81	0.76
2	8.10	0.72	7.73	0.78
3	7.96	0.73	7.42	0.83
4	**8.04**	**0.75**	**7.35**	**0.83**
5	8.03	0.73	7.49	0.81

Training Loss on Nations dataset for margin 1 to 5 for TransR



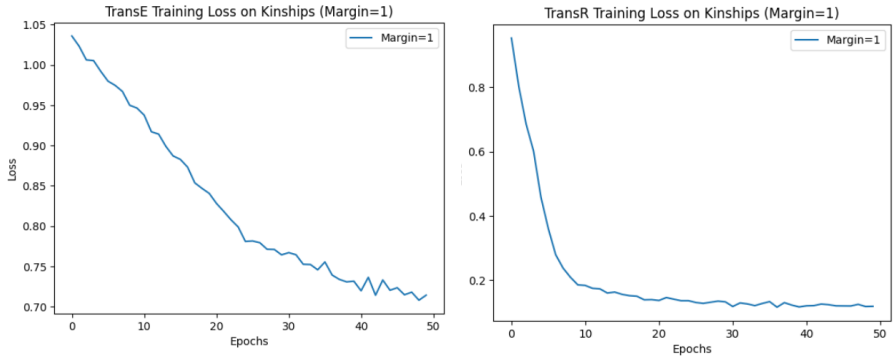
TransE Performance on Nations Dataset

Margin	Mean Rank (Head)	Hits@10 (Head)	Mean Rank (Tail)	Hits@10 (Tail)
1	48.05	0.10	41.80	0.14

TransR Performance on Nations Dataset

Margin	Mean Rank (Head)	Hits@10 (Head)	Mean Rank (Tail)	Hits@10 (Tail)
1	19.91	0.30	16.83	0.39

Training Loss on Kinships dataset for margin 1 on TransE and Trans R



Performance on Nations Dataset:

TransE achieves good accuracy for the simpler political relations in the Nations dataset. With margin 3, TransE attains relatively low Mean Ranks (7.82 for head, 6.97 for tail) and high Hits@10 values (0.70 for head, 0.81 for tail), showing that it can effectively capture these simpler relationships. However, as the margin increases, Hits@10 tends to fluctuate, indicating a slight loss in ranking quality at higher margins.

TransR performs best with margin 4 in the Nations dataset, consistently achieving higher Hits@10 scores and lower Mean Ranks. This demonstrates that TransR's ability to embed entities into relation-specific spaces provides an advantage when modeling political relations with some complexity, such as intergovernmental organizations or embassies. TransR also maintains higher Hits@10 scores (0.75 for head, 0.83 for tail), showing its effectiveness in ranking accuracy.

Performance on Kinships Dataset:

TransE shows limited performance on Kinships, with higher Mean Ranks (48.05 for head, 41.80 for tail) and low Hits@10 (0.10 for head, 0.14 for tail). This suggests that TransE struggles with the intricate, context-dependent relationships present in the Kinships dataset, where a single linear transformation is insufficient.

TransR, on the other hand, performs significantly better on Kinships, with much lower Mean Ranks (19.91 for head, 16.83 for tail) and higher Hits@10 (0.30 for head, 0.39 for tail). TransR's use of relation-specific projections enables it to better capture the nuanced relationships in the dataset, leading to improved accuracy and ranking.

General Comparison:

Simplicity vs. Complexity: TransE is suitable for datasets with simpler relational structures, such as the Nations dataset, where the model can effectively represent connections through translational embeddings. However, TransR excels in more complex scenarios due to its relation-specific transformations, handling intricate relationships like those in the Kinships dataset more effectively.

Ranking Metrics: Across both datasets, TransR typically achieves better Mean Ranks and Hits@10 than TransE, particularly at optimal margins. This indicates that TransR provides more consistent ranking quality, especially when faced with datasets containing varied relationships.

Optimal Margin Dependence: Both models demonstrate varying performance based on margin, but TransR generally shows less sensitivity to changes in margin compared to TransE, suggesting its robustness in diverse relationship contexts.

Similar Fact Retrieval Results

Embedding Dimension: 100

The similarity scores represent closeness of triples with the validation triples based on their embeddings.

Top 5 Similar Triples for Validation Triples

Validation Triple 1: ('brazil', 'commonbloc1', 'india')

Top 5 similar triples using TransE embeddings with scores:

Triple: ('netherlands', 'commonbloc1', 'india') Similarity Score: 0.8679

Triple: ('usa', 'commonbloc1', 'india') Similarity Score: 0.8464

Triple: ('uk', 'commonbloc1', 'india') Similarity Score: 0.8286

Triple: ('brazil', 'commonbloc1', 'egypt') Similarity Score: 0.7987

Triple: ('brazil', 'commonbloc1', 'israel') Similarity Score: 0.7481

Validation Triple 2: ('burma', 'intergovorgs3', 'indonesia')

Top 5 similar triples using TransE embeddings with scores:

Triple: ('burma', 'commonbloc2', 'indonesia'), Similarity Score: 0.8648

Triple: ('burma', 'relngo', 'indonesia'), Similarity Score: 0.8605

Triple: ('burma', 'conferences', 'indonesia'), Similarity Score: 0.8407

Triple: ('burma', 'relexports', 'indonesia'), Similarity Score: 0.8173

Triple: ('burma', 'intergovorgs3', 'usa'), Similarity Score: 0.8134

Validation Triple 3: ('china', 'accusation', 'uk')

Top 5 similar triples using TransE embeddings with scores:

Triple: ('china', 'accusation', 'usa'), Similarity Score: 0.9337

Triple: ('china', 'accusation', 'india'), Similarity Score: 0.9139

Triple: ('china', 'accusation', 'ussr'), Similarity Score: 0.8679

Triple: ('india', 'accusation', 'uk'), Similarity Score: 0.7599

Triple: ('china', 'relbooktranslations', 'uk'), Similarity Score: 0.6956

Validation Triple 4: ('cuba', 'reldiplomacy', 'china')

Top 5 similar triples using TransE embeddings with scores:

Triple: ('cuba', 'reldiplomacy', 'ussr'), Similarity Score: 0.8116

Triple: ('cuba', 'reldiplomacy', 'uk'), Similarity Score: 0.7791

Triple: ('cuba', 'commonbloc2', 'china'), Similarity Score: 0.7705

Triple: ('cuba', 'reldiplomacy', 'poland'), Similarity Score: 0.7434

Triple: ('cuba', 'reldiplomacy', 'indonesia'), Similarity Score: 0.7298

Validation Triple 5: ('egypt', 'embassy', 'uk')

Top 5 similar triples using TransE embeddings with scores:

Triple: ('egypt', 'embassy', 'usa'), Similarity Score: 0.9228

Triple: ('egypt', 'embassy', 'india'), Similarity Score: 0.9000

Triple: ('egypt', 'embassy', 'netherlands'), Similarity Score: 0.8941

Triple: ('egypt', 'embassy', 'poland'), Similarity Score: 0.8658

Triple: ('egypt', 'embassy', 'brazil'), Similarity Score: 0.8345

Validation Triple 1: ('brazil', 'commonbloc1', 'india')

Top 5 similar triples using TransR embeddings with scores:

Triple: ('brazil', 'commonbloc1', 'jordan'),	Similarity Score: 0.8202
Triple: ('brazil', 'commonbloc1', 'burma'),	Similarity Score: 0.8179
Triple: ('brazil', 'commonbloc1', 'egypt'),	Similarity Score: 0.7949
Triple: ('israel', 'commonbloc1', 'usa'),	Similarity Score: 0.7748
Triple: ('brazil', 'commonbloc1', 'israel'),	Similarity Score: 0.7731

Validation Triple 2: ('burma', 'intergovorgs3', 'indonesia')

Top 5 similar triples using TransR embeddings with scores:

Triple: ('burma', 'intergovorgs3', 'usa'),	Similarity Score: 0.8705
Triple: ('burma', 'intergovorgs3', 'egypt'),	Similarity Score: 0.8131
Triple: ('burma', 'intergovorgs3', 'uk'),	Similarity Score: 0.7699
Triple: ('burma', 'intergovorgs3', 'india'),	Similarity Score: 0.7628
Triple: ('burma', 'intergovorgs3', 'brazil'),	Similarity Score: 0.7473

Validation Triple 3: ('china', 'accusation', 'uk')

Top 5 similar triples using TransR embeddings with scores:

Triple: ('china', 'accusation', 'india'),	Similarity Score: 0.8887
Triple: ('india', 'accusation', 'uk'),	Similarity Score: 0.8675
Triple: ('indonesia', 'accusation', 'china'),	Similarity Score: 0.6233
Triple: ('ussr', 'accusation', 'china'),	Similarity Score: 0.6040
Triple: ('usa', 'accusation', 'china'),	Similarity Score: 0.5977

Validation Triple 4: ('cuba', 'reldiplomacy', 'china')

Top 5 similar triples using TransR embeddings with scores:

Triple: ('cuba', 'reldiplomacy', 'indonesia'),	Similarity Score: 0.9301
Triple: ('cuba', 'reldiplomacy', 'ussr'),	Similarity Score: 0.9295
Triple: ('ussr', 'reldiplomacy', 'china'),	Similarity Score: 0.9169
Triple: ('poland', 'reldiplomacy', 'china'),	Similarity Score: 0.9144
Triple: ('cuba', 'reldiplomacy', 'poland'),	Similarity Score: 0.9129

Validation Triple 5: ('egypt', 'embassy', 'uk')

Top 5 similar triples using TransR embeddings with scores:

Triple: ('egypt', 'embassy', 'poland'),	Similarity Score: 0.8599
Triple: ('ussr', 'embassy', 'uk'),	Similarity Score: 0.8282
Triple: ('egypt', 'embassy', 'usa'),	Similarity Score: 0.7979
Triple: ('netherlands', 'embassy', 'uk'),	Similarity Score: 0.7826
Triple: ('poland', 'embassy', 'uk'),	Similarity Score: 0.7685